GOAL DIRECTION ORIENTATION, GENERALIZED TURNING HABIT AND GOAL GRADIENT AS FACTORS IN MAZE LEARNING IN THE RAT[1]

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According to the principle of goal gradient elaborated by Hull (1), the goal reaction becomes more strongly conditioned to the stimuli preceding the goal in time or space than to the stimuli which are more remote. It is deduced from this hypothesis that blind alleys near the goal should be eliminated sooner or entered less frequently than those more remote. That is to say, according to this principle there should occur a backward elimination of blind alleys in a maze. A superficial examination of figure 1 showing the difficulty, expressed as percentages of total errors, of of the blind alleys of three similar multiple-T mazes might cause one to question in some degree the validity of the concept of goal gradient as an important factor in the learning of this type of maze. However, the maze is not a simple learning situation and it is accordingly reasonable to suppose that numerous factors enter to determine the difficulty of the blind alleys and hence to obscure the goal gradient.

In this paper the author will attempt to show that two factors other than serial position determine in part the difficulty of blind alleys; that these factors can be isolated by analyzing the error curves of blind alleys possessing certain characteristics in common; and that those alleys which are free from such disturbing factors actually do show a definite difficulty gradient.

Records from four groups of animals learning multiple-T type mazes under varying conditions will be considered in this report. Table 1 gives a summary of the conditions governing the learning

[1] From a paper read before the American Psychological Association, September, 1932.

of the four groups. The patterns of the three mazes employed are given in figure 2. Mazes B and C are easily recognized as

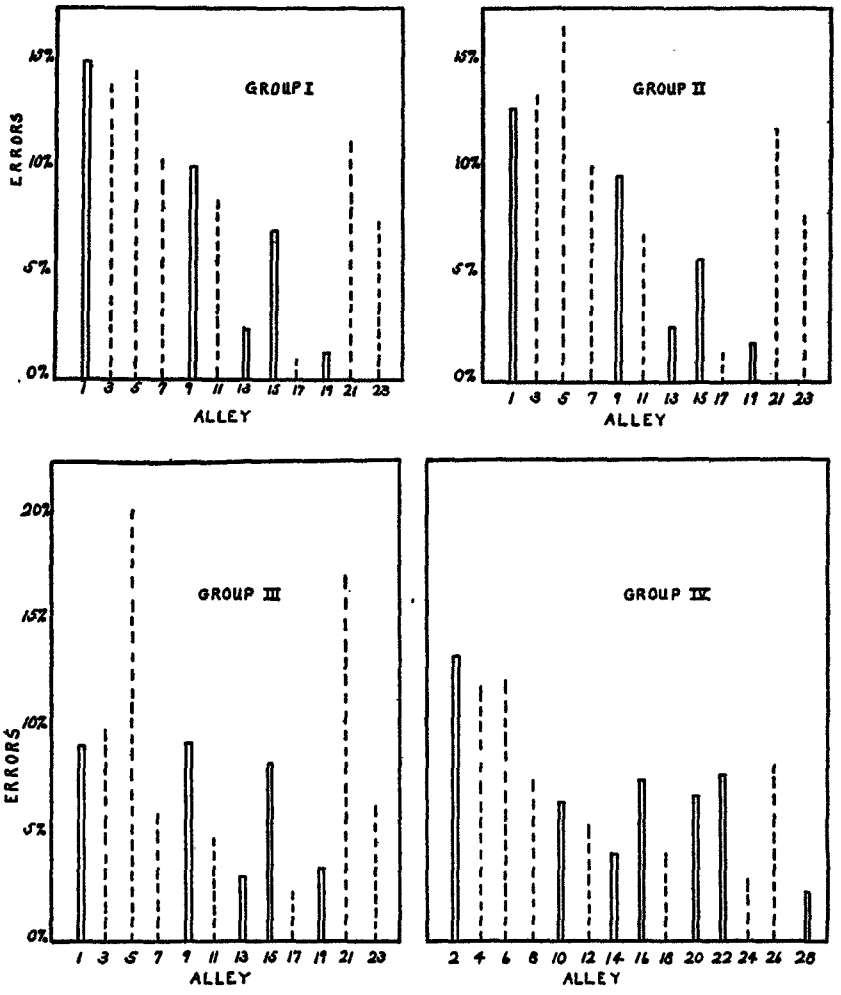


FIG. 1. SHOWING THE DIFFICULTY OF EACH BLIND ALLEY AS THE PERCENTAGE THAT THE NUMBER OF ENTRIES IS OF THE TOTAL NUMBER OF ERRORS THE SOLID-LINE BARS REPRESENT COMPARABLE BLIND ALLEYS

copies of the multiple-T mazes of Stone (4) and Tolman (5). Maze A is a slightly modified copy of the Stone maze and has been described elsewhere in detail (2) (3).

First examine Maze A in figure 2. Observe that the twelve blind alleys of this maze are numbered 1, 3, 5, etc. to 23. Notice further that each blind can be classified as goalward pointing or non-goalward pointing. Blind alleys 3, 5, 7, 11, and 21 are goalward pointing and the others are non-goalward. Blind alleys 7 and 21 are not so definitely goalward as the others but since separate analysis of them proved that they behave exactly as the others, they have been included in the goalward group. Now

TABLE 1

GROUP	N	MAZE	EXPERIMENTER	MOTIVATION	DISTRIBUTION OF TRIALS
I	38	A	Ruch	Escape water	1 trial per day, then 2 per day
II	19	A	Ruch	Escape water	1 trial per day
III	54	B	Stone*	Hunger food	1 trial per day
IV	28	C	Anderson*	Hunger food	1 trial per day

* Thanks are expressed to Professor Calvin P. Stone and to Mr. Edward E. Anderson for permission to use maze records from their files.

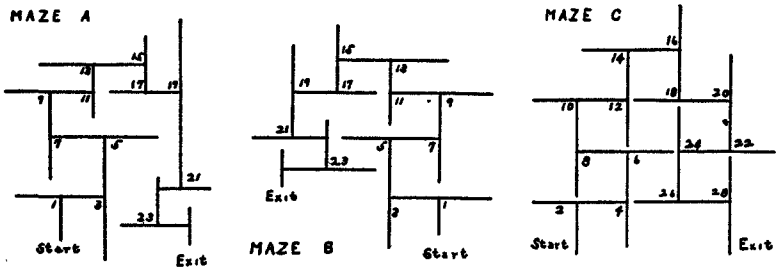


FIG. 2. SHOWING THE FLOOR PLANS OF THE MAZES

note that the true path of Maze A contains 14 turns of which 6 are to the *left* and 8 to the *right*. Note also that 3 of the 5 goalward blind alleys are *right* turns. The fact that the true path of the maze contains more *right* than *left* turns leads one to search for a possible generalized habit of turning to the right. Later we shall see that this is actually found.

It will be seen at a glance that Maze B is essentially a mirror image of the one just discussed. Of the 14 turns in its true path,

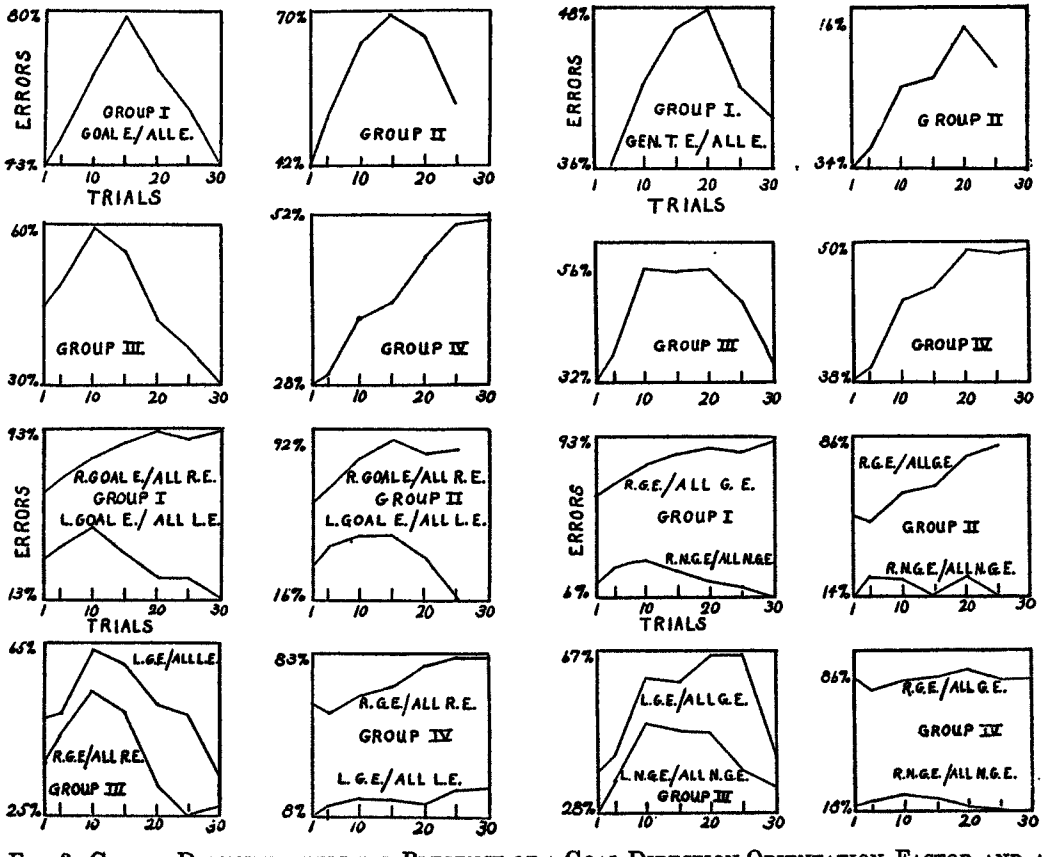


FIG. 3. CURVES DEMONSTRATING THE PRESENCE OF A GOAL DIRECTION ORIENTATION FACTOR AND A GENERALIZED TURNING HABIT

six are to the *right* and the remaining eight including the final one are to the *left*. There are but four goalward blind alleys, numbers 5, 7, 11, and 21. Two of them are *right* and two of them *left* turns.

Let us now examine Maze C. Its true path contains six *left* turns and eight *right* including the final one. Observe that there are four goalward blind alleys and that only one of them is a *left* turn.

The four curves in the upper left corner of figure 3 demonstrate the presence of a goal direction orientation factor which operates to increase the frequency of entry into goalward pointing blind alleys and hence to conceal the presence of a simple goal gradient as based on all of the blinds of the mazes.

These four curves picture the increase with practice in the relative frequency of entry into the goalward pointing blind alleys. Each point on the curves represents the number of entries into the goalward blinds divided by the total number of errors made during each segment of the trial series. In all four groups we see that there is a progressive tendency for the goalward pointing blind alleys to be entered relatively more frequently than the non-goalward. This fact is the more significant because of the varied conditions of motivation and distribution of trials which characterized the training programs of the four groups. Group IV was still making a large number of errors on the thirtieth trial whereas the other groups had practically eliminated all errors. Consequently, the entire curve of Group IV corresponds to the first part of the other curves and is thus in complete agreement with them.

In view of the inescapable presence of a factor making for a difficulty of goalward blind alleys disproportionate to their serial position in the maze, it would be confusing to attempt to compare goalward with non-goalward blind alleys in establishing the presence or absence of a goal gradient. That is to say, the difficulty of a given blind alley is determined in part by its nearness to the goal and in part by whether it points in the goalward or non-goalward direction. It might be expected that all goalward blind alleys would show one form of goal gradient and that all non-goalward blind alleys would show another.

It has already been noted that each maze pattern presents a preponderance of turns in one direction. Since this is true one might expect that in Maze A, for example, which has in its true path more *right* than *left* turns a generalized right-turning habit to arise causing the right-turning blind alleys to be entered more frequently than the left-turning ones. To test the validity of this conjecture the blind alleys of the three mazes were separated into two groups accordingly as they represent turns in the same direction as the majority of the true path turns or in a direction opposite to the majority of the true path turns. If the number of the errors made in the blind alleys opening to the same hand as the majority of the true-path alleys is divided by the total number of errors made during each segment of the trial series, the resultant values would indicate the presence or absence of a generalized turning habit. The four curves given in the upper right quarter of figure 3 show that such a generalized turning habit does appear during the earlier part of the learning process. In the case groups A, B, and D the relative frequency of right errors increases from the first trial while in the case data from Maze C the relative frequency of the left-turn errors increases. The explanation might be that the rats form a generalized turning habit by virtue of a greater number of true path turns in one direction than in the other.

A further complication arises. In the case of Mazes A and D the final turn as well as the majority of the true-path turns are to the right. A comparable situation exists also in maze C. It is therefore impossible to separate the action of this factor from that of generalized turning habit.

In the lower left corner of figure 3 are four sets of curves showing the relative frequency of entry into goalward blind alleys as determined for the right-turning and the left-turning alleys separately. Scrutiny of these curves leaves little doubt of the existence in its own right of a goal orientation factor. In every instance an initial rise is observed.

In corresponding fashion the four sets of curves given in the lower right corner of figure 3 show that a generalized turning habit is present in a slight degree at least. Those goalward blind

alleys which open to the same hand as the majority of the true path turns are entered with progressively greater relative frequency as learning proceeds through its early stages. A similar treatment of the non-goalward errors gives a similar result in three of the four groups studied. Although the evidence for a generalized turning habit is not so strong as that for the existence of a goal orientation factor, it is, nevertheless, clear enough to show the validity of basing the determination of the presence or absence of a goal gradient on those blind alleys which are entirely homogeneous in that they represent non-goalward and contra-general turns.

After this laborious analysis, it is possible to present in simple fashion the data indicating the reality of a goal gradient in the performance of the four groups studied.

Return to figure 1. In these graphs the heighth of the solid-line bars represents the frequency of entry into each of the non-goalward, contra-general, blind alleys as a percentage of the total number of errors made during the entire trial series. The evidence from the two upper figures is unmistakably in favor of a steep goal gradient, alley 13 being the only one out of place and it only slightly so. The other two groups do not give such clear evidence of a goal gradient but such as it is the evidence points in that direction.

If this analysis has any value it lies as much in indicating the complexity of the factors present in the learning of a moderately simple maze as in showing the validity of the goal gradient hypothesis. It does, however, indicate that a goal gradient might be expected to appear more clearly in mazes which do not present an opportunity for goal-direction orientation or generalized turning habits.[2]

[2] The reader is urged to compare the results reported here with those given by K. W. Spence in: "The order of eliminating blinds in maze learning by the rat." Jour. Comp. Psychol., 1932, xiv, 9-27. It is interesting to note that these analyses, performed independently, lead to the same general conclusions. This paper is of further interest in that it discusses previous investigations of the problem of the order of blind alley elimination to which reference has been here omitted for the sake of brevity.

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